

LOW-COST RESCUE DRONE

Engineering Project Overview | UAV Control & Embedded Systems

Period	July 2024 – January 2025	Team	2 members; I was project lead
Lab	Atal Tinkering Lab, Aristo Public School	Final controller	KK2.1.5
Focus	Electronics, embedded systems, UAV control	Outcome	Working prototype + flight demonstration

PROJECT IN BRIEF

I began this project to understand how a quadcopter works by building and testing one myself. The project progressed from an ESP32 + MPU6050 concept, to an Arduino + MPU6050 experimental stage, and finally to a KK2.1.5-based working prototype. I led the technical development, hardware integration, assembly, testing, troubleshooting and documentation.

PROTOTYPE EVOLUTION

- **ESP32 + MPU6050:** embedded control, sensing and wireless-control exploration.
- **Arduino + MPU6050:** stabilization, calibration, ESC control and radio integration.
- **KK2.1.5:** system integration, configuration, calibration and testing.

ENGINEERING WORK

- Arduino-based experimental control and MPU6050 sensor integration.
- Sensor calibration, ESC control and radio integration.
- Wiring and integration of receiver, controller, ESCs, motors and power system.
- KK2.1.5 configuration, calibration, testing and troubleshooting.
- Project planning, documentation, teamwork and presentation.

WHAT I LEARNED

- Engineering is iterative.
- Breaking a large system into smaller subsystems helps troubleshooting.
- Reliable hardware integration depends on careful wiring, calibration and repeated testing.
- Building with real components increased my interest in microcontrollers, electronics, embedded systems and control.

CURRENT RESEARCH DIRECTION

The project motivated me to develop a deeper understanding of UAV guidance, navigation, flight control, aerial robotics and autonomous systems, and to learn how a manually operated working prototype can be developed toward more autonomous and research-oriented aerial systems.

TECHNICAL SCOPE & LIMITATIONS

This was an educational working prototype, not a commercial rescue system. Autonomous navigation, AI-based detection, obstacle avoidance and thermal imaging were not implemented. Formal measurements such as flight time, altitude, range and payload were not recorded.

RECOGNITION & EVIDENCE

The project was demonstrated at school and presented to students. The portfolio records recognition from school leadership and participation in Innovators Day 2024.